Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 9 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 9 : SLO â€“ 1 | Queue Implementation Using Linked List

■ I. Enqueue Operation â€” Insert an Element VAL

• Step 1: Allocate memory for the new node and name it as PTR
• Step 2: SET PTR -> DATA = VAL
• Step 3: IF FRONT = NULL
• SET FRONT = REAR = PTR
• SET FRONT -> NEXT = REAR -> NEXT = NULL
• ELSE
• SET REAR -> NEXT = PTR
• SET REAR = PTR
• SET REAR -> NEXT = NULL
• [END OF IF]
• Step 4: END

■ II. Dequeue Operation â€” Delete an Element

• Step 1: IF FRONT = NULL
• Write UNDERFLOW
• Go to Step 5
• [END OF IF]
• Step 2: SET PTR = FRONT
• Step 3: SET FRONT = FRONT -> NEXT
• Step 4: FREE PTR
• Step 5: END